F- Test distribution :-

Formula :- $\bar{x}_1 = \frac{\Sigma x_1}{n}$ $\bar{x}_2 = \frac{\Sigma x_2}{n}$

$$S_1^2 = \frac{\Sigma(x_1 - \bar{x}_1)^2}{n_1 - 1} \qquad S_2^2 = \frac{\Sigma(x_2 - \bar{x}_2)^2}{n_2 - 1}$$

$$F_c = \frac{S_2^2}{S_1^2}$$

① The following table is the number of units Produced Per day by two worker's A & B. For some days can we say that worker is more stable than worker. B? Use 5% level of significance ($F_{0.05} = 4.88$)

worker A	40	30	38	41	38	35	-	-
worker B	39	38	41	33	32	49	49	34

Soln :-

worker A (x_1)	$(x_1-\bar{x}_1)$	$(x_1-\bar{x}_1)^2$
40	3	9
30	-7	49
38	1	1
41	4	16
38	1	1
35	-2	4

$\Sigma(x_1 - \bar{x}_1)^2 = 80$

$$x_1 = \frac{\Sigma x_1}{n} = \frac{222}{6} = 37$$

$$S_1^2 = \frac{\Sigma(x_1-\bar{x}_1)^2}{n_1 - 1} = \frac{80}{6-1} = \frac{80}{5} = 16.$$

DATE

x_2

worker B	$(x_2-\bar{x}_2)$	$(x_2-\bar{x}_2)^2$
39	-0.37	0.13
38	-1.37	1.87
41	1.63	1.63
33	-6.37	40.57
32	-7.37	54.31
49	9.63	92.73
49	9.63	92.73
34	-5.37	28.83

$\Sigma x_2 = 315$ $\Sigma(x_2-\bar{x}_2)^2 = 312.8$

$$\bar{x}_2 = \frac{315}{8} = 39.37$$

$$S_2^2 = \frac{\Sigma(x_2-\bar{x}_2)^2}{n_2-1} = \frac{312.8}{8-1} = \frac{312.8}{7} = 44.68$$

$$F_c = \frac{S_2^2}{S_1^2} = \frac{44.68}{\cancel{2}16} = 2.79$$

Hypothesis is accepted $(F = 4.88)$

$\underline{2.79 > 4.88}$

T- test

Formula :- ~~let~~ $|t| = \left| \frac{\bar{x} - \mu}{s} \cdot \sqrt{n} \right|$

$$\bar{x} = \frac{\Sigma x}{n} \qquad s^2 = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2$$

Problem :-

A certain stomach administrator provides each patient with 12 tablets per week, routed in a fixed order in ~~+~~ two blood pressure conditions. The numbers of tablets taken over 12 patients are. 5, 2, 8, −1, 3, 0, 6, −2, 1, 5, 0, 4 Can it be concluded that the stomach remedy is effective in controlling ~~bo~~ blood. Pressure at the 5%. los ?

[$t_{0.05} = 2.201$]

Sol^n :- $\bar{x} = \frac{\Sigma x}{n} = \frac{5+2+8-1+3+0+6-2+1+5+0+4}{12} = \frac{31}{12} = 2.58$

$\bar{x} = 2.58$ //

$$s^2 = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2$$

$$= \frac{1}{12-1} \Big[(5-2.58)^2 + (2-2.58)^2 + (8-2.58)^2 + (-1-2.58)^2 + (3-2.58)^2 + (-2-2.58)^2 + (5-2.58)^2 + (1-2.58)^2 + (6-2.58)^2 + (4-2.58)^2 \Big]$$

$$s^2 = \frac{104.91}{11} \Rightarrow s^2 = \sqrt{9.5378} \Rightarrow \boxed{s = 3.088}$$

$\sqrt{n} = \sqrt{12} = 3.464$

$\mu \to$ did't given so Put as Zero.

$|t| = \left| \frac{\bar{x} - \mu}{s} \cdot \sqrt{n} \right|$

$= \left| \frac{2.58 - 0}{3.088} \cdot 3.464 \right|$

$|t| = 2.894$

$t_{0.05} = (2.201) < 2.894.$

$\therefore$ Hypothesis is rejected

② Consider the Sample Consisting of Nine ~~10~~ number 45, 47, 50, 52, 48, 47, 49, 53, 51. the Sample is drawn from a Population whose mean is 47.5. Find whether the Sample mean ~~is it~~ differ significantly from the Population mean at 5% los. ($t_{0.05} = 2.31$)

Chi-Square [Binomial distribution]

(1) A **Formula :-** $P(x) = {}^{n}C_{x}\, p^{x} q^{n-x}$

$$F(x) = N.\, P(x).$$

Problem :-

A set of five similar coins is tossed 320 times and the result is.

No. of Heads	0	1	2	3	4	5
frequencies.	6	27	72	112	71	32

Test the hypothesis that the data follow a binomial distribution ($\chi^2_{0.05} = 11.07$)

Soln :- Given $V = n-1 = 6-1 = 5.$

$\chi^2 = 11.07.$

$n = 5.$

Binomial distribution is :- $P(x) = {}^{n}C_{x}\, p^{x} q^{n-x}$

P = Probability of getting head $= \frac{1}{2}.$

$q = 1-P = 1-\frac{1}{2} = 1/2.$

$F(x) = N.P(x)$, $\boxed{N = 320}$

$$F(x) = \cancel{8}\, N \cdot P(x) \qquad \therefore P(x) \Rightarrow {}^{n}C_{x}\, p^{x} q^{n-x}$$

$$F(x) = 320 \cdot {}^{5}C_{x}\, (0.5)^{x} \cdot (0.5)^{5-x}$$

where $x=0$, $F(0) = 10$

$x=1$, $F(1) = 50$

$x=2$, $F(2) = 100$

$x=3$, $F(3) = 100$

$x=4$, $F(4) = 50$

$x=5$, $F(5) = 10$

Experimentaly we got data.

O_i	6	27	72	112	71	32
E_i	10	50	100	100	50	10

– observed data is given in Problem

$$\chi^2 = \frac{\Sigma (O_i - E_i)^2}{E_i}$$

$$\chi^2 = \frac{(6-10)^2}{10} + \frac{(25-50)^2}{50} + \frac{(72-100)^2}{100} + \frac{(112-100)^2}{100} + \frac{(71-50)^2}{50\cancel{00}} + \frac{(32-10)^2}{10}$$

$$\chi^2 = \frac{7868}{100} = 78.68 \Rightarrow \boxed{\chi^2 = 78.68}$$

$$\chi^2_{0.05} = 11.07.$$

$$\chi^2 > \chi^2_{0.05}$$

$\therefore$ Hypothesis is rejected.

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